Isabel Esther Rodríguez Alvarez

Backend & AI Agent Engineer

Python · FastAPI · Microsoft Azure · RAG · Multi-Agent · MCP

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Summary

Software engineer with **5 years of experience** building **Python backends** and **production AI agents**. At La Nave, one of Cuba's most popular ride-hailing apps (**10,000+ users**), I build FastAPI services and event-driven Azure architectures and shipped **semantic and hybrid search** to production. As a freelancer, I design and build production AI agents **from scratch** for healthcare, real estate and hospitality: **multi-agent systems**, RAG, **MCP** and function calling.

Experience

Backend & AI Developer · La Nave · Remote

Oct 2024 – Present

Ride-hailing app connecting passengers with nearby drivers; 10,000+ users.

- Build and maintain the backend **REST APIs** in Python with **FastAPI**, with **JWT/OAuth2** authentication and **Redis** caching.
- Implemented app-to-backend communication with **Azure SignalR** (real time) and **Service Bus** (event-driven architecture) on Azure Functions and App Services.
- **Redesigned the address search** with **semantic and hybrid search** on Azure AI Search over **160,000+ addresses and places**: it suggests results in real time as users type, without blocking the event channel (async processing with Service Bus).
- **La Nave official store (in production):** built a Node.js/TypeScript plugin for semantic and hybrid product search and an **Azure AI Foundry agent** that automatically validates the product catalog.
- Design, model and manage **PostgreSQL** (Azure Database for PostgreSQL) and **Cosmos DB** (MongoDB API) databases.
- Automated deployments with **Docker**, Azure Container Registry and **Azure DevOps CI/CD** pipelines; monitor services with Application Insights.

AI Agent Developer · Freelance · Remote

2024 – Present

Agents built from scratch: requirements with clients (English and Spanish), architecture, development, Azure deployment and support.

- **VitalIA (healthcare): multi-agent medical board** (~9 specialists plus critic and consensus agents) producing clinical reports with **deterministically calculated confidence**, 8 report agents running in parallel on FHIR data, and real-time medical chat. 3 microservices: FastAPI, Functions, Service Bus, SignalR, AI Foundry.
- **Xuxa · ED Bienes Raíces (Panama): 24/7 real estate advisor on WhatsApp, Telegram and web** that searches CRM listings, **captures leads automatically** and delegates legal questions to a sub-agent with **hybrid RAG** citing law and article. 3 custom MCP servers; DEV/PROD environments.
- **Sage · restaurant chain (Panama):** inventory and purchasing agent with **64 custom MCP tools**, 5 roles with approvals and separation of duties (**105/105 permission tests passing**), photo-evidence receiving and scheduled restock alerts.

Backend Developer · Datalisoft · Remote

Sep 2021 – Sep 2024

Cuban software development company.

- Built backend APIs and services in Python with **FastAPI** and **Django**.
- Built **custom AI assistants** for clients across industries with **Azure OpenAI** and the Azure AI SDK, tailored to each business's data and processes.
- Developed **React** and **Streamlit** interfaces integrated with backend services.

Programming Instructor (social service) · UCI

Sep 2021 – Sep 2023

- Taught OOP with Java, Data Structures I & II and Web Programming; labs with Spring and Django.

Technical Skills

Languages & frontend: Python, TypeScript, Java, C++ · React, Streamlit
Backend: FastAPI, Django, Spring, Node.js, REST APIs, JWT/OAuth2, Redis, event-driven architecture
AI Agents & LLMs: Azure OpenAI (GPT), Claude, Llama, Mistral · LangChain, Microsoft Agent Framework, OpenAI Agents SDK, Azure AI Foundry · RAG, embeddings, vector and hybrid search, function calling, memory, multi-agent, MCP, streaming
Microsoft Azure: Functions, Service Bus, SignalR, App Services, AI Search, AI Foundry, Blob Storage, Azure Files, Container Registry
Data & DevOps: PostgreSQL, Cosmos DB / MongoDB, Redis · Docker, Azure DevOps (CI/CD), Git, Application Insights, OpenTelemetry

Education

B.Sc. Computer Science Engineering · UCI, Cuba · **Gold Degree** (highest honors) *2021*

Languages

Spanish: native · **English:** B2, daily professional use (meetings, requirements and documentation with clients)